SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

COMPUTER SCIENCE AND ENGINEERING PROGRAMME

CSA3717 – SOFTWARE TESTING

EXPERIMENT 13

ONLINE EXAMINATION SYSTEM – POSITIVE TEST SCENARIOS AND TEST CASES

Aim:

To Write a case study to identify and analyze the requirements of an Online Examination System and prepare positive test scenarios and test cases for the application.

1. Problem Description

An online examination system allows registered students to authenticate, view assigned examinations, read instructions, answer questions, save responses and submit within the permitted examination window.

The Online Examination System is expected to provide reliable, secure and user-friendly services. The testing team verifies the stated requirements using valid and invalid inputs, records actual results, and reports any deviation from the expected behaviour.

2. Functional Requirements

1. Only registered students may access their assigned examination.
2. The examination must be available only during the configured time window.
3. Instructions must be displayed before the examination starts.
4. The system must save selected answers reliably.
5. The timer must reflect the configured examination duration.
6. Students must not submit after the permitted deadline unless an extension is authorized.
7. Submission must be recorded once and produce a confirmation.
8. Results must be visible only when released by an authorized role.

3. Positive Test Scenarios and Test Cases

Positive testing checks whether the application performs the required operation when valid data and correct user actions are supplied.

Test Case ID	Test Scenario	Test Input / Action	Expected Result	Status
PT01	Student login	Valid student credentials	Student dashboard is displayed.	Pass
PT02	View assigned exam	Open an exam assigned to the student	Exam title, date and duration are displayed.	Pass
PT03	Read instructions	Open exam instructions	Instructions are displayed before starting.	Pass
PT04	Start examination	Start within the permitted time window	Question paper opens and timer starts.	Pass
PT05	Answer objective question	Select one valid option	Selected answer is saved.	Pass
PT06	Navigate questions	Move to next and previous questions	Question navigation works correctly.	Pass
PT07	Submit examination	Submit before the deadline	Submission confirmation is displayed.	Pass
PT08	View released result	Open result after authorized release	Score and permitted feedback are displayed.	Pass

5. Detailed Execution of a Failed Test Case

Test Case ID: PT08

Precondition: A registered user has a valid account and the required service is available.

Test Steps: 1. Sign in with valid credentials. 2. Open the relevant module. 3. Enter valid data. 4. Submit the operation.

Expected Result: The application completes the operation and displays a success confirmation.

Actual Result: The operation completes successfully in this illustrative execution.

Conclusion: The positive test passes when actual behaviour matches the expected result.

Conclusion

The test cases prepared for the Online Examination System provide coverage of its key functional requirements. Executing them helps identify functional gaps, improve error handling and increase confidence in the quality of the application.